# Securing understanding of sequences

## Task A: Continuing sequences from a rule

1) For each two starting numbers: continue the sequence with a constant additive pattern, then a constant multiplicative pattern. Describe each of your sequences in words.

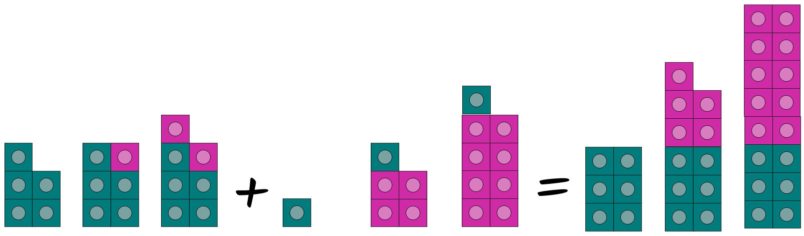
| | Additive pattern | Multiplicative pattern |
|---|---|---|
| a) Sequence | 1, 3, 5, 7, 9, … | 1, 3, 9, 27, 81 … |
| Description | Start on 1 then add 2 each time | Start on 1 then multiply by 3 each time |
| b) Sequence | 0.1, 1, 1.9, 2.8, 3.7, … | 0.1, 1, 10, 100, 1000, … |
| Description | Start on 0.1 then add 0.9 each time | Start on 0.1 then multiply by 10 each time |
| c) Sequence | −16, 8, 32, 56, 80, … | −16, 8, −4, 2, −1, … |
| Description | Start on (−16) then add 24 each time | Start on −16 then multiply by $-\frac{1}{2}$ |
| d) Sequence | 0.5, 1.5, 2.5, 3.5, 4.5, … | 0.5, 1.5, 4.5, 13.5, 40.5, … |
| Description | Start on 0.5 then add 1 | Start on 0.5 then multiply by 3 |
| e) Sequence | $\frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{4}{8} = \frac{1}{2}, \frac{5}{8}$ … | $\frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2$ … |
| Description | Start on $\frac{1}{8}$ and add $\frac{1}{8}$ | Start on $\frac{1}{8}$ and multiply by 2 |
| f) Sequence | 25, 10, −5, −20, −35, … | 25, 10, 4, $\frac{8}{5}$, $\frac{16}{25}$, … (25, 10, 4, 1.6, 0.64, …) |
| Description | Start on 25 then subtract 15 | Start at 25 then multiply by $\frac{2}{5}$ |

## Task B: Representations of sequences

Izzy has built these patterns using blocks. Which two patterns could she combine to make a sequence which

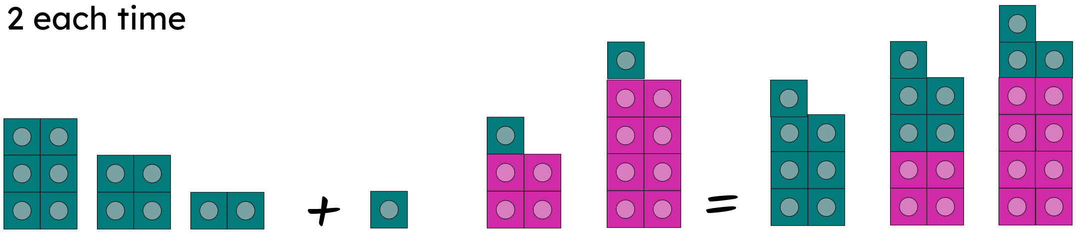
1) Increases by 5 each time

*A and D*



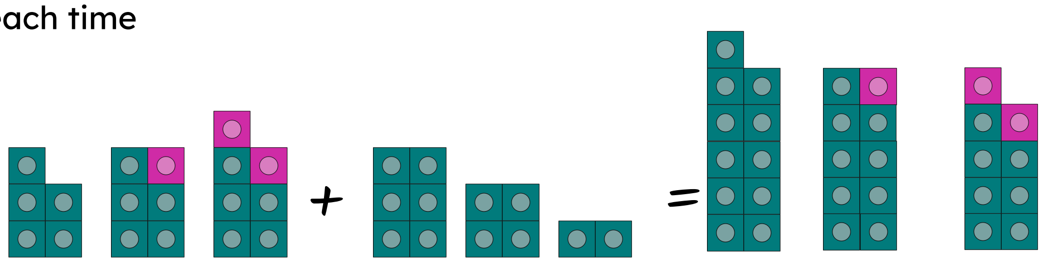
2) Increases by 2 each time

*B and D*

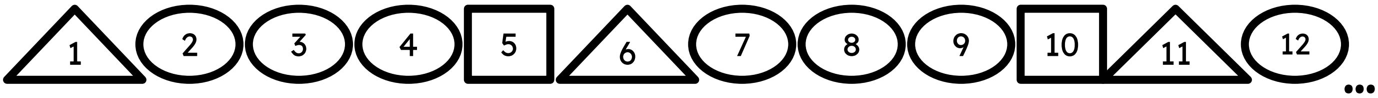


3) Decreases by 1 each time

*A and B*



4) Laura and Andeep are making a repeating pattern from different shaped counters




a) What shape counter will the number 25 be?

*Square as all multiples of 5 are squares*

b) Laura puts down the counters 144, 145 and 146, what shapes are they?

*Oval, square, triangle as squares are multiples of 5 so 145 must be square*

c) Andeep puts down three counters in a row which are all oval shapes. They are all greater than 50 but less than 60, what numbers could they be?

*52, 53, 54 or 57, 58, 59.*

d) Laura and Andeep run out of counters when they put down the counter 178. How many of each type did they use in total